In 1809 Malus found that the *intensity of polarized light transmitted through a polarizer is proportional to the square of cosine of the angle between the plane of polarization of the light and the transmisson axis of the polarizer.*

This statement is known as Malus' law.

If unpolarized light of intensity I_o is incident on a polarizer, plane polarized light of intensity $I_o/2$ is transmitted by it. Let us denote $I_o/2$ by I_1. Let this plane polarized light pass through an analyzer. The intensity of the light transmitted through the analyzer is given by

$$I = E_1^2 \cos^2\theta = I_1 \cos^2\theta = \frac{1}{2} I_0 \cos^2\theta \quad \ldots (20.10)$$

Light transmitted through the analyzer at specific settings are as follows.

Case (*i*): If $\theta = 0°$	axes parallel	$I = I_1 = \frac{1}{2} I_0$
Case (*ii*): If $\theta = 90°$	axes perpendicular	$I = 0$
Case (*i*): If $\theta = 180°$	axes parallel	$I = I_1 = \frac{1}{2} I_0$
Case (*i*): If $\theta = 270°$	axes perpendicular	$I = 0$

Thus, we obtain two positions of maximum intensity and two positions of zero intensity when we rotate the axis of the analyzer with respect to that of the polarizer.

20.10. ANISOTROPIC CRYSTALS

When a light beam is incident on an *isotropic medium* such as a glass slab, it refracts as a single ray. An optically isotropic material is one in which the index of refraction is the same in all directions. Glass, water and air are examples of isotropic materials. The atoms in a crystal are arranged in a regular periodic manner. If the arrangement of atoms differ in different directions within a crystal, then the physical properties vary with the direction. The thermal conductivity, electrical conductivity, velocity of light and hence refractive index etc properties depend on the crystallographic direction along which the property is measured. Then we say that the crystal is **anisotropic**. In such anisotropic crystals the force of interaction between the electron cloud and the lattice is different in different crystallographic directions. The natural frequency of the electron cloud is likewise dependent on the direction in which the electrons are caused to vibrate by the incident light wave. This results in different velocities in different directions and the index of refraction is different in different directions within the crystal.

The anisotropic crystals are divided into two classes: uniaxial and biaxial crystals. In case of **uniaxial crystals**, one of the refracted rays is an ordinary ray and the other is an extraordinary ray. In **biaxial crystals** both the refracted rays are extraordinary rays. Calcite, tourmaline and quartz are examples of uniaxial crystals whereas mica, topaz and aragonite are examples of biaxial crystals.

20.10.1 CALCITE CRYSTAL

Calcite is a common naturally occurring substance. Both marble and limestone are made up of many small calcite crystals bonded together. A large crystal of calcite is colourless and transparent. It was at one time found in great quantities in Iceland and hence it is also known as **Iceland spar**. Naturally occurring calcite crystals (Fig.20.24*a*) has rhombohedral cleavage which means it breaks into blocks with parallelogram-shaped faces. It is bounded by six faces (Fig.20.24*b*), each of which is a parallelogram with angles equal to 101°55'and 78°5'. The rhombohedron has only two corners A and H where all the face angles are obtuse (101°55'). These two corners appear as the *blunt corners* of the crystal. At the rest of six corners there is one obtuse angle and two acute angles.

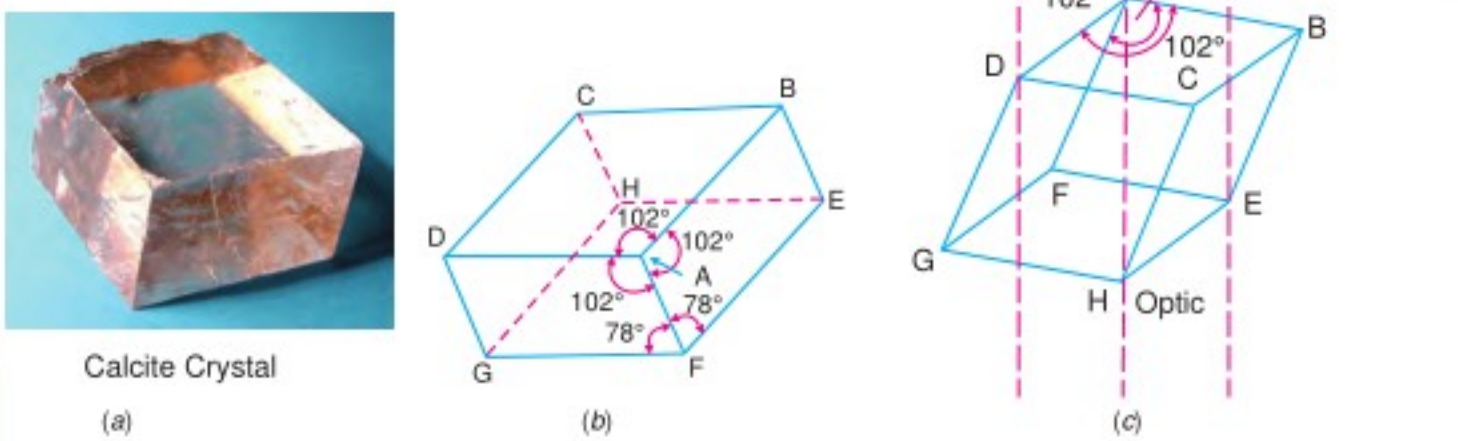


Fig. 20.24

20.10.2 OPTIC AXIS

A line bisecting any one of the blunt corners (A or H) and making equal angles with each of the three edges meeting there, is the **optic axis** (see Fig.20.24 *c*). In fact any line parallel to this line is also an optic axis. Thus, the optic axis is a direction and not a specific line in the crystal. Hence an optic axis can be drawn through every point in the crystal, that is, any line parallel to the line above will also be the optic axis. It is to be noted that the optic axis is not obtained by joining the two blunt corners. Only in a special case, when the three edges of the crystal are equal, the line joining the two blunt corners A and H coincide with the crystallographic axis of the crystal and it gives the direction of the optic axis. The optic axis is actually the axis of symmetry of the crystal. A ray of light propagating along optic axis does not suffer double refraction, because the structure of the crystal is symmetric about that direction.

The optic axis is the direction in a uniaxial crystal along which the e-ray and the o-ray travel with the *same* speed and consequently double refraction *does not* take place along this direction. The corresponding refractive index is the refractive index for ordinary light, say μ_o.

20.10.3 PRINCIPAL SECTION

A plane containing the optic axis and perpendicular to a pair of opposite faces of the crystal is called the **principal section** of the crystal for that pair of faces (Fig.20.25 *a*). Thus, there are three principal sections passing through any point within the crystal, one corresponding to each pair of opposite faces. A principal section always cuts surfaces of calcite crystal in a parallelogram having angles 71° and 109° (see Fig.20.25 *b*). Fig.20.25 (*c*) shows a face of the crystal in which the end-view

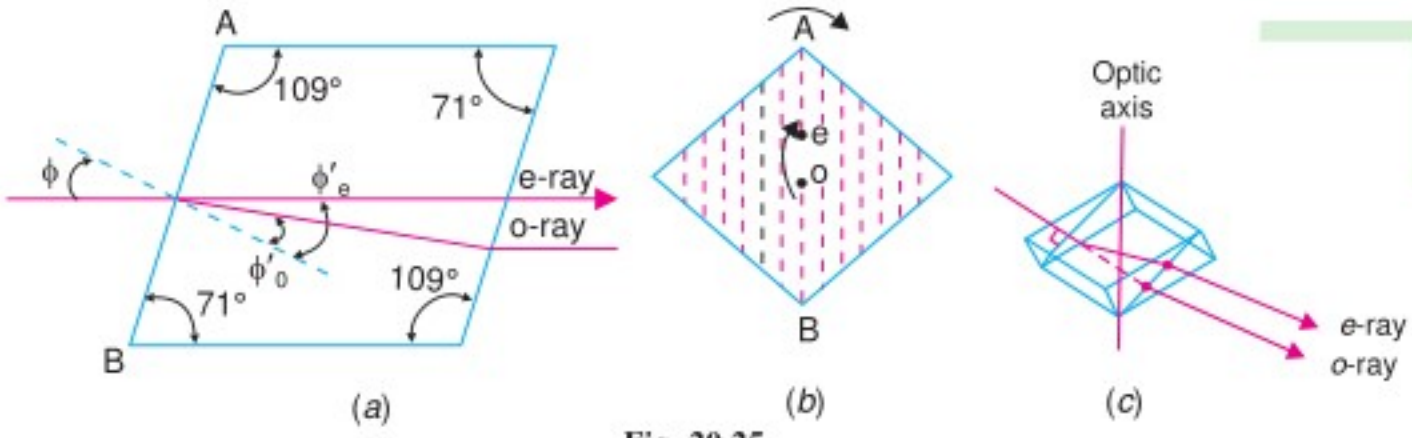


Fig. 20.25

of the principal section AB (of Fig.20.25 b) is shown by the dotted line AB. The lines parallel to AB represent the end-views of other principal sections parallel to AB with in the crystal.

20.10.4 PRINCIPAL PLANE

Defining principal section is not enough to understand the directions of vibrations for the o-ray and e-rays. Hence, two more planes are defined as principal plane for the o-ray and the principal plane for the e-ray. The plane containing the optic axis and the o-ray is called the *principal plane of the o-ray* and the plane containing the optic axis and the e-ray is called the *principal plane of the e-ray*. The directions of vibrations in the o-ray and e-ray can be understood with reference to these planes. In general, the two principal planes do not coincide. Under the particular case, when the plane of incidence is the principal section of the crystal, then the principal planes of o- and e-rays and the principal section of the crystal coincide.

20.11. DOUBLE REFRACTION IN CALCITE CRYSTAL

Fig.20.25 (*a*) shows a principal section of calcite crystal. A ray of light is incident on the face AB of the crystal and it travels along the principal section. The ray is split into two rays, namely o-ray (fast ray) and e-ray (slow ray). The o-ray travels through the crystal without deviation while the e-ray is refracted at some angle. As the opposite faces of the crystal are parallel, the rays emerge out parallel to the incident ray. Within the crystal the o-ray *always* lies in the plane of incidence whereas e-ray does not lie in the plane of incidence. e- ray lies in the plane of incidence only when the plane of incidence is a principal section.

If a mark (dot or cross) is made on a paper and then the calcite crystal (AB face) is placed on it, two images are seen through the crystal, as illustrated in Fig.20.25 (*b*). The images are produced by the o-ray and e-ray. The intensities of the images are lesser than that of the original mark. The line joining them lies in the principal section. If now the crystal is rotated slowly about an axis passing through the o-image, the e-image moves round in a circle while the o-image remains stationary. It shows that the velocity of propagation of o-ray is the same in all directions, while that of e-ray changes with direction. O-ray obeys the laws of refraction and the e-ray does not follow the ordinary laws of refraction.

The e-ray and o-ray are linearly polarized. The e-ray has its vibrations (i.e., the optical vector) *parallel* to the principal section whereas the vibrations (optical vector) in o-ray are *perpendicular* to the principal section, as indicated in Fig.20.19. The vibration directions can be established by examining the rays through a polarizer. As the polarizer is held in the path of the rays and rotated slowly, the intensity of one of the images, say the o-image, increase while that of the e-image decreases. In one position, the intensity of the o-image will be a maximum while the e-image is extinguished. Further rotation through 90° from this particular position, causes the o-image to disappear and e-image intensity to become a maximum. It proves that the e- and o-rays are linearly polarized in mutually perpendicular directions.

The o-ray travels with the same velocity in all directions in the crystal whereas the e-ray travels with different velocities in different directions. Therefore, refractive index corresponding to o-ray is a constant and is designated by μ_o. The refractive index corresponding to e-ray varies and is denoted by μ_e. The difference between the refractive indices is known as the amount of double refraction or birefringence. Thus,

$$\Delta_\mu = \mu_e - \mu_o \qquad \ldots (20.11)$$

20.11.1 HUYGENS' EXPLANATION OF DOUBLE REFRACTION

In order to explain the double refraction exhibited by anisotropic crystals, Huygens postulated that the incident light excites two separate wavelets within the crystal, one spherical wavelet associated

with the ordinary waves and one ellipsoidal wavelet associated with the extraordinary waves. For example, the plane wavefront, in Fig.20.26, incident normally on the crystal surface generates spherical wavelets as well as ellipsoidal wavelets. The spherical wavelets propagate equally quickly in all directions. The wave surface corresponding to o-ray is therefore spherical. The tangent to these waves lies straight ahead and, by successive application of the principle, the plane wave propagates straight ahead with speed υ. The ellipsoidal wavelets propagate at different speeds in different directions. The wave surface corresponding to e-ray is therefore an ellipsoid of revolution about the optic axis. The common tangent to these ellipsoids after a little time is the new wavefront. The line from the point of generation of each ellipsoid to the tangent point on that ellipsoid is off at an angle and defines the direction of travel of the extraordinary wavefronts. The wavefronts are not perpendicular to their direction of travel.

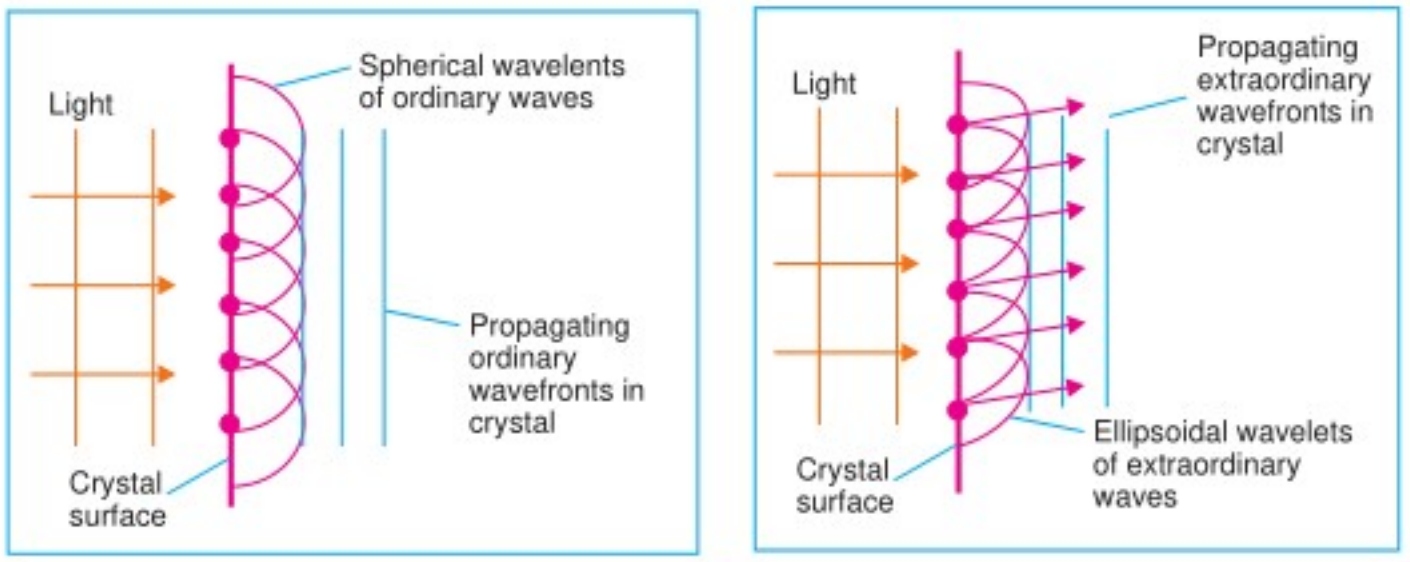


Fig. 20.26

The two wave surfaces touch each other at the two points where they are intersected by the optic axis. As light propagates through the crystal, the two wave surfaces travel in different directions in the crystal. Ultimately, two refracted rays emerge from the crystal.

20.11.2 ORDINARY AND EXTRA-ORDINARY RAYS

We now compare the properties of o- and e-rays:

(*i*) o-ray obeys the laws of refraction and the e-ray does not follow the ordinary laws of refraction.

(*ii*) Both o-ray and e-ray are plane polarized. They are polarized in mutually perpendicular planes. The electric vector of o-ray vibrates perpendicular to the principal section of o-ray while the vibrations of e-ray take place parallel to the principal section of e-ray.

(*iii*) O-ray travels with the same speed in all directions within the crystal. The e-ray travels with different speeds along different directions in the crystal. However, the speed of e-ray will be equal to that of o-ray along the optic axis direction.

(*iv*) Because o-ray travels with the same velocity in all directions, the refractive index corresponding to it has a constant value. On the other hand, the refractive index for e-ray varies from direction to direction. The principal refractive index for o-ray is defined as follows:

$$\mu_0 = \frac{c}{\upsilon_0} = \frac{\text{velocity of light in a vacuum}}{\text{velocity of o-ray in the crystal}} \qquad \ldots (20.12)$$

The principal refractive index for e-ray in ***positive crystals*** is defined as follows:

$$\mu_o = \frac{c}{(\upsilon_e)_{min}} = \frac{\text{velocity of light in a vacuum}}{\text{minimum velocity of e-ray in the crystal}} \quad \ldots (20.13)$$

The principal refractive index for e-ray in ***negative crystals*** is defined as follows:

$$\mu_e = \frac{c}{(\upsilon_e)_{max}} = \frac{\text{velocity of light in a vacuum}}{\text{maximum velocity of e-ray in the crystal}} \quad \ldots (20.14)$$

(*v*) When natural light is incident on an anisotropic crystal at an angle to the optic axis, it splits into o-and e-rays, which travel in *different directions* with *different velocities* (20.27 *a*).

When natural light is incident in a direction perpendicular to the optic axis, o-ray and e-ray propagate in the *same direction* in the crystal but with *different velocities*, as shown in Fig.20.27 (*b*). In a negative crystal e-ray leads o-ray and in case of a positive crystal o-ray leads e-ray.

When natural light is incident on the crystal in a direction parallel to the optic axis, it does not split into two rays. The o- and e- rays travel in the *same direction with the same velocity*, as shown in Fig.20.27 (*c*).

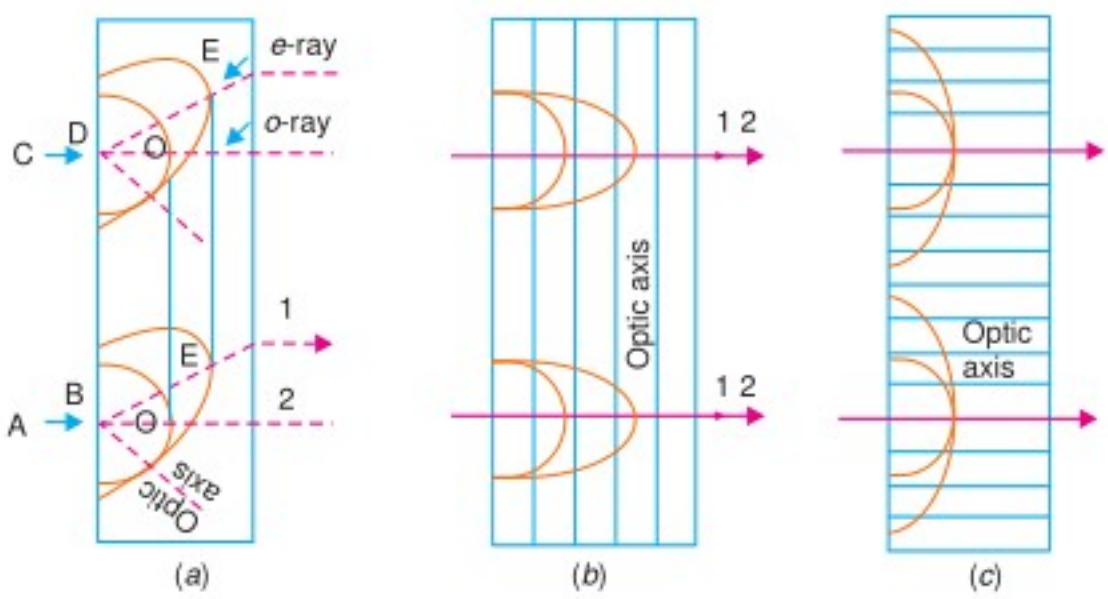


Fig. 20.27

(*vi*) The distinction of o-ray and e-ray exists only within the crystal. Once they emerge from the crystal, they travel with the same velocity. The rays outside the crystal differ only in their direction of travel and plane of polarization. The designation of o- ray and e-ray has no meaning outside the crystal.

20.11.3 POSITIVE AND NEGATIVE CRYSTALS

Because of two different wave fronts, two different types of uniaxial crystals exist. In one type of crystals, the spherical wave front of o-ray is enclosed by the ellipsoidal wave front of e-ray. Such crystals are known as **negative crystals**. They are called negative crystals because the refractive index corresponding to the e-ray is less than that corresponding to o-ray. Calcite crystal is an example of negative type crystals. In the other case, the extraordinary wave front lies within the ordinary wave front and such crystals are called **positive crystals**. They are positive because the refractive index for the extraordinary ray is greater than that of o-ray. Quartz crystal is an example of positive crystals.

We compare here the characteristics of the positive and negative crystals.

(*i*) In positive uniaxial crystals, the ellipsoid of revolution corrrsponding to the e-ray is totally contained within the sphere corresponding to the o-ray.

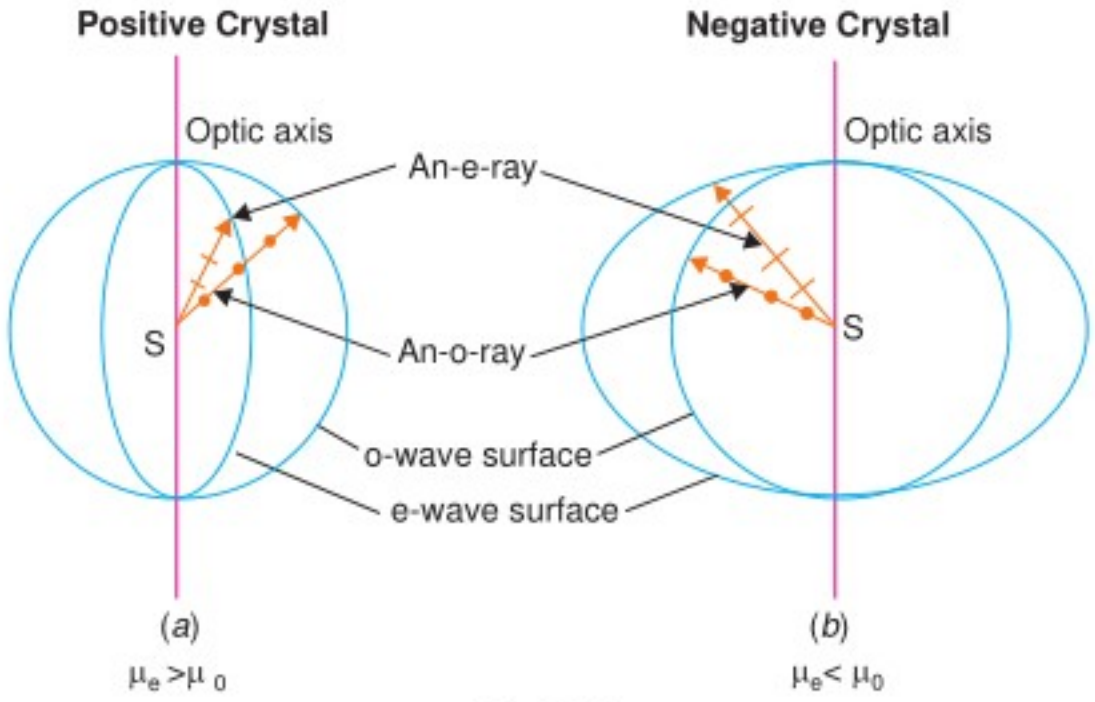


Fig. 20.28

In negative uniaxial crystals, the ellipsoid of revolution for e-ray lies completely outside the sphere corresponding to o-ray. The two cases are depicted in Fig. 20.28.

(*ii*) In positive crystals the e-ray velocity has a maximum value along the optic axis and a minimum value in a direction perpendicular to the optic axis.

On the other hand, in negative crystals the velocity of e-ray has a minimum value parallel to the optic axis and a maximum value in a direction perpendicular to the optic axis.

(*iii*) In positive crystals, e-ray travels slower than o-ray in all directions except along the optic axis.

$$v_e = v_o \quad — \quad \text{parallel to optic axis}$$
$$v_e < v_o \quad — \quad \text{other directions}$$

In negative crystals, o-ray travels slower than e-ray in all directions except along the optic axis.

$$v_e = v_o \quad — \quad \text{parallel to optic axis}$$
$$v_e > v_o \quad — \quad \text{other directions}$$

(*iv*) In positive crystals the principal refractive index for e-ray is larger than the principal refractive index for o-ray.

$$\mu_e > \mu_o$$

In negative crystals the principal refractive index for o-ray is larger than the principal refractive index for e-ray.

$$\mu_e < \mu_o$$

(*v*) **Birefringence** or amount of double refraction of a crystal is defined as

$$\Delta\mu = \mu_e - \mu_o \qquad \ldots (20.15\ a)$$

As $\mu_e > \mu_o$ in these crystals, $\Delta\mu$ is a positive quantity for positive crystals.

$\Delta\mu$ is a negative quantity for negative crystals as $\mu_e < \mu_o$ in these crystals and the birefringence of a negative crystal is given by

$$\Delta\mu = \mu_o - \mu_e \qquad \ldots (20.15\ b)$$

(*v*) When **elliptically polarized light** is incident on the polarizer, the transmitted light varies in intensity from a maximum value I_{max} to a minimum value I_{min} on rotation of the polarizer (Fig.20.31). occurs when the transmission axis of the polarizer coincides with the semi-major axis of the ellipse and occurs when the transmission axis coincides with the semi-minor axis of the ellipse.

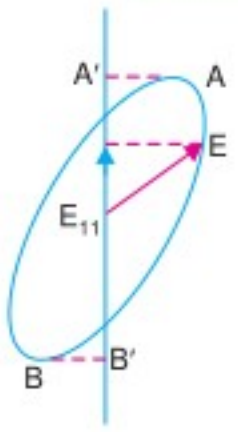


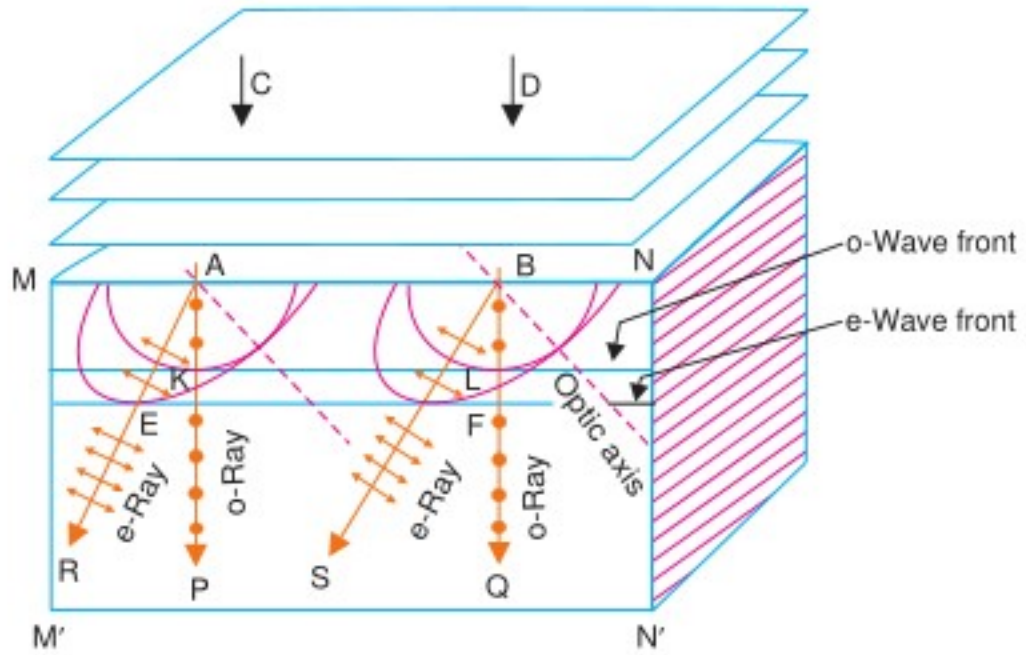
Fig. 20.31

20.14. HUYGENS' CONSTRUCTION OF WAVEFRONTS:

A thinner rectangular cross-section of a crystal can be cut from a bigger crystal in three different ways. It can be cut in such a way that the optic axis lies (*i*) inclined to the refracting face or (*ii*) parallel to the refracting face or (*iii*) perpendicular to the refracting face. The path of o-ray and e-ray within a uniaxial crystal can be determined using Huygens' principle of secondary wavelets. We take the example of a negative crystal for the purpose of tracing the paths of o-ray and e-ray.

Case.1. Optic axis inclined to the refracting edge:

Let MNN′M′ represent the calcite (negative) crystal. Let CD represent the monochromatic plane wave front incident normal to the crystal surface. The optic axis is ***inclined to the refracting edge.*** According to Huygens the phenomenon of double refraction involves two types of propagation of light waves. The wave known as o-wave has a spherical wave front whereas the e-wave has an ellipsoidal wave front. The two wave fronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.



Unpolarized light beam is incident at normal on a calcite crystal slice. Optic axis is inclined to the incident ray. Huygens wavelets showing double refraction, and difference in velocities of *o*-ray and *e*-ray.

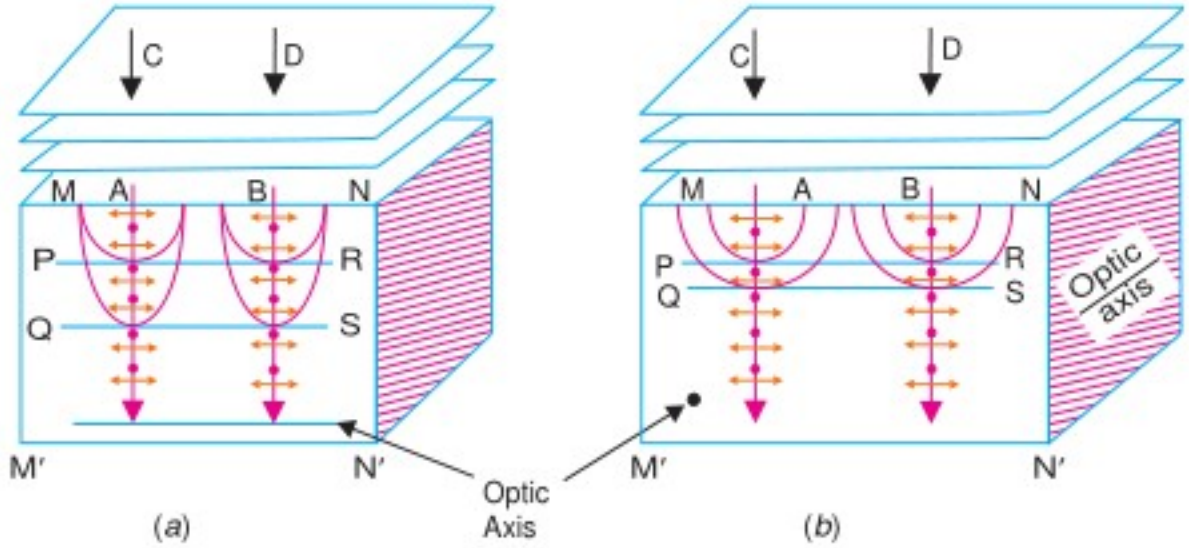
Fig. 20.32

Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wave front becomes a source of secondary disturbance. The points A and B are chosen for the purpose of illustration.

According to Huygens principle the points A and B produce elliptical and spherical wavelets. The position of the two wave fronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly, a circle of the same radius is drawn with B as centre. The ellipsoidal wave front can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. The circle and the ellipse touch each other, as required, along the optic axis. If we now draw the common tangents to the secondary wavelets, they represent the plane wave fronts corresponding to the two rays. Thus KL is the tangent to the spherical wavelets and EF is the tangent to the ellipsoidal wave fronts. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to K and B to L, we obtain the direction of propagation of o-ray. Similarly, the lines AP and BQ show the direction of propagation of e-ray. We find that in this case the o- ray and e-ray travel along different directions with different velocities.

Case 2(*a*): Optic axis in the plane of incidence and parallel to the refracting edge:

Let MNN′M′ represent the refracting face of the calcite (negative) crystal. Let CD represent the monochromatic plane wavefront incident normal to the crystal surface. Let the optic axis be ***parallel*** to the refracting edge MN and lie in the plane of incidence. See Fig.20.33 (a). According to Huygens the phenomenon of double refraction involves propagation of light waves of two types. The wave known as o-wave has a spherical wavefront whereas the e-wave has an ellipsoidal wavefront. The two wavefronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.



Unpolarized light beam incident normally on a calcite slab at right angles to the optic axis. Double refraction does not occur but *o*-ray and *e*-ray travel with different velocities in the crystal. (*a*) Optic axis in the plane of incidence and parallel to the crystal surface; (*b*) Optic axis perpendicular to the plane of incidence and parallel to the crystal surface.

Fig. 20.33

Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wavefront becomes a source of secondary disturbance. The points A and B are chosen for the purpose of illustration. According to Huygens' principle the points A and B produce elliptical and spherical wavelets. The two wavelets touch each other along AM and BN, which is the direction of optic axis, as required. The wave fronts at any instant can be drawn as tangential surfaces to the two wavelets. Note that the two wave fronts here are parallel to each other. The position of the two wavefronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly,

a circle of the same radius is drawn with B as centre. The ellipsoidal wavefront can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. PR is the tangent to the spherical wavelet and therefore represents the refracted wavefront for the o-ray. Similarly, QS represents the refracted wavefront for e-ray. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to P and B to R, we obtain the direction of propagation of o-ray. By joining A to Q and B to S we get the direction of propagation of e-ray. We find that in this case the o- ray and e-ray travel along the same direction but have different velocities.

Case 2 (*b*): Optic axis perpendicular to the plane of incidence and parallel to the refracting edge:

Let MN be the refracting edge of the calcite (negative) crystal. Let CD represent the monochromatic plane wave front incident normal to the crystal surface. The optic axis is ***parallel to the refracting edge but is perpendicular to the plane of incidence*** (Fig.20.33 b). According to Huygens the phenomenon of double refraction involves two types of propagation of light waves. The wave known as o-wave has a spherical wave front whereas the e-wave has an ellipsoidal wave front. The two wave fronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.

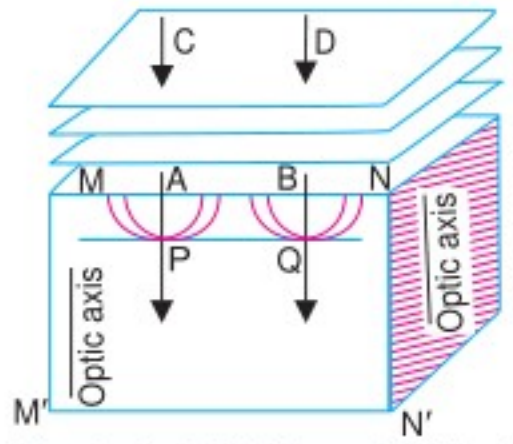
Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wave front becomes a source of secondary disturbance. The points A and B are chosen for the purpose of illustration. According to Huygens principle the points A and B produce elliptical and spherical wavelets. Since the spherical and ellipsoidal wavelets touch each other along a line perpendicular to the plane of incidence, the wavelets appear spherical in the plane of incidence. The position of the two wave fronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly, a circle of the same radius is drawn with B as centre. The ellipsoidal wave front can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. The circle and the ellipse touch each other, as required, along the optic axis, which is perpendicular to the surface MN. If we now draw the common tangents to the secondary wavelets, they represent the plane wavefronts corresponding to the two rays. Thus PR is the common tangent to the spherical wavelets and therefore represents the refracted wave front for the o-ray. Similarly, QS represents the refracted wave front for e-ray. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to P and B to R, we obtain the direction of propagation of o-ray. By joining A to Q and B to S we get the direction of propagation of e-ray. We find that in this case the o- ray and e-ray travel along the same direction but have different velocities.

Case 3 : Optic axis perpendicular to the refracting edge and lying in the plane of incidence:

Let MNN′M′ represent the calcite (negative) crystal. Let CD represent the monochromatic plane wave front incident normal to the crystal surface. The optic axis is ***perpendicular to the refracting edge and lies in the plane of incidence***. According to Huygens the phenomenon of double refraction involves two types of propagation of light waves. The wave known as o-wave has a spherical wave front whereas the e-wave has an ellipsoidal wave front. The two wave fronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.

Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wavefront becomes a new point source of light. The points A and B are chosen for the purpose of illustration. According

to Huygens' principle the points A and B produce elliptical and spherical wavelets. Since the two wavelets must touch each other along the optic axis, they touch at points P and Q (Fig.20.34). Note that the section of the ellipsoid lies outside the section of the circle. The minor axis of the ellipse and the radius of the circle are equal. The position of the two wave fronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly, a circle of the same radius is drawn with B as centre. The ellipsoidal wave front can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. The circle and the ellipse touch each other, as required, along the optic axis, which is perpendicular to the refracting edge MN. If we now draw the common tangents to the secondary wavelets, they represent the plane wavefronts corresponding to the two rays. Thus PQ is the common tangent to the spherical wavelets as well as the ellipsoidal wave fronts. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to P and B to Q, we obtain the direction of propagation of o-ray as well as e-ray. We find that in this case the o- ray and e-ray travel along the same direction with the same velocity.



Unpolarized light beam incident normally on a calcite slab parallel to the optic axis. The o-ray and e-ray travel along the same velocity.

Fig. 20.34

20.11. EXPERIMENTAL DETERMINATION OF PRINCIPAL REFRACTIVE INDICES

For determining the refractive index for the extraordinary ray a calcite crystal is cut in the form of a prism with the optic axis perpendicular to the refracting edge of the prism and perpendicular to the base BC (Fig.20.35). It can also be cut with the optic axis parallel to the refracting edge of the prism. The prism is placed on the spectrometer table and is adjusted for the minimum deviation position for the extraordinary rays. The angle of minimum deviation δ_m is determined and the principal refractive index for the extraordinary ray is calculated from the relation (20.16a).

$$\mu_e = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}} \qquad \ldots (20.16\ a)$$

For a given wavelength, the ordinary and the extraordinary rays are separated while passing through the prism. Therefore, the angle of minimum deviation δ_m' for the ordinary ray can be measured and thus its refractive index can be calculated from the relation (20.16b).

$$\mu_0 = \frac{\sin\left(\frac{A+\delta_m'}{2}\right)}{\sin\frac{A}{2}} \qquad \ldots (20.16\ b)$$

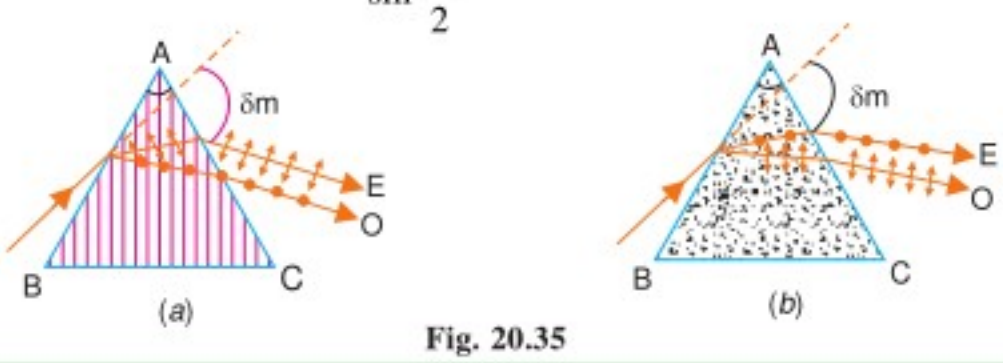


Fig. 20.35